

Discrete Mathematics: Mid-Semester Master Guide

Comprehensive Coverage: Sets, Relations, & Functions — Theory, Proofs & Solved Problems

Target: Comprehensive Exam Prep | Coverage: Definitions, Operational Laws, Proofs, Solved Medium/Hard Problems & Applications

1. Sets, Operations, Cardinality & Countability

Fundamental Definitions

Set: A well-defined collection of distinct objects (elements). Formally, $x \in A$ indicates that x is an element of set A .

Power Set $P(A)$: The set of all subsets of A . If $|A| = n$, then $|P(A)| = 2^n$.

Review of Basic Set Operations & Laws

For sets A, B, C within a universal set U :

- **Union ($A \cup B$):** $\{x \mid x \in A \vee x \in B\}$
- **Intersection ($A \cap B$):** $\{x \mid x \in A \wedge x \in B\}$
- **Set Difference ($A \setminus B$ or $A - B$):** $\{x \mid x \in A \wedge x \notin B\}$
- **Symmetric Difference ($A \oplus B$):** $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$
- **Absolute Complement (A^c or A'):** $U - A$

Core Set Identities & Laws

- **De Morgan's Laws:** $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$
- **Distributive Laws:** $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- **Inclusion-Exclusion Principle:** For three finite sets:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

Cardinality & Countability

Cardinality ($|A|$): A measure of the "number of elements" of a set.

- **Countable Set:** A set A is countable if it is finite or has the same cardinality as the set of natural numbers $\mathbb{N}$ (i.e., there exists a bijection $f: \mathbb{N} \rightarrow A$). If infinite and countable, its cardinality is denoted by $\aleph_0$ (Aleph-null).
- **Uncountable Set:** An infinite set that cannot be put into one-to-one correspondence with $\mathbb{N}$ (e.g., $\mathbb{R}$, $(0, 1)$, $P(\mathbb{N})$). Its cardinality is often denoted by c (continuum, $2^{\aleph_0}$).

Key Proofs & Exam Questions (Sets)

PROOF 1: Cantor's Diagonal Argument CLASSIC PROOF

Theorem: The set of real numbers in the open interval $(0, 1)$ is uncountable.

Proof:

We proceed by contradiction. Assume $(0, 1)$ is countable. Then its elements can be listed as a sequence $r_1, r_2, r_3, \dots$

Write each real number in its non-terminating decimal representation:

$$r_1 = 0 . d_{11} d_{12} d_{13} d_{14} \dots$$

$$r_2 = 0 . d_{21} d_{22} d_{23} d_{24} \dots$$

$$r_3 = 0 . d_{31} d_{32} d_{33} d_{34} \dots$$

Construct a number $x = 0 . x_1 x_2 x_3 \dots$ in $(0, 1)$ by defining the k -th digit as:

$$x_k = 4 \text{ if } d_{kk} \neq 4, \text{ else } x_k = 5$$

By construction, x differs from r_1 in the 1st decimal digit, from r_2 in the 2nd digit, and in general, from r_k in the k -th digit. Thus $x \neq r_k$ for all $k \in \mathbb{Z}^+$. This contradicts the assumption that the list contains all real numbers in $(0, 1)$. Hence, $(0, 1)$ is uncountable. ■

PROBLEM 1.1: Inclusion-Exclusion Application MEDIUM

In a group of 100 computer science students, 45 know Python, 40 know Java, and 50 know C++. 15 know Python and Java, 20 know Java and C++, and 18 know Python and C++. If 10 students know all three languages, how many students know **none** of these languages?

Solution:

Let P, J, C be the sets of students knowing Python, Java, and C++ respectively.

By Principle of Inclusion-Exclusion:

$$|P \cup J \cup C| = |P| + |J| + |C| - (|P \cap J| + |J \cap C| + |P \cap C|) + |P \cap J \cap C|$$

Substitute values: $|P \cup J \cup C| = 45 + 40 + 50 - (15 + 20 + 18) + 10 = 135 - 53 + 10 = 92$.

Number of students knowing none = $|U| - |P \cup J \cup C| = 100 - 92 = 8$.

PROBLEM 1.2: Countability of Rational Numbers HARD

Prove that the set of all positive rational numbers $\mathbb{Q}^+$ is countable.

Proof:

Arrange all rational numbers p/q in an infinite two-dimensional grid where row i represents numerator $p = i$ and column j represents denominator $q = j$.

Traverse the array along diagonals: $(1/1)$, $(1/2, 2/1)$, $(1/3, 2/2, 3/1)$, $(1/4, 2/3, 3/2, 4/1)$, ...

Assign a unique natural number index to each entry encountered, skipping fractions that are not in lowest terms (e.g., skip $2/2$ because it equals $1/1$). This forms an explicit bijection $f: \mathbb{N} \rightarrow \mathbb{Q}^+$. Thus, $\mathbb{Q}^+$ is countably infinite. ■

2. Relations, Types, Operations & Recurrence Relations

Binary Relation

A binary relation R from set A to set B is a subset of the Cartesian product $A \times B$. We write $a R b$ for $(a, b) \in R$.

Types of Binary Relations on a Set A

Property	Formal Condition ($\forall a, b, c \in A$)	Visual / Matrix Representation
Reflexive	$(a, a) \in R$	Main diagonal of adjacency matrix contains all 1s.
Irreflexive	$(a, a) \notin R$	Main diagonal of adjacency matrix contains all 0s.
Symmetric	$(a, b) \in R \Rightarrow (b, a) \in R$	Adjacency matrix is symmetric ($M = M^T$).
Antisymmetric	$((a, b) \in R \wedge (b, a) \in R) \Rightarrow a = b$	No non-diagonal pair of symmetric 1s exists.
Transitive	$((a, b) \in R \wedge (b, c) \in R) \Rightarrow (a, c) \in R$	If path of length 2 exists, direct edge must exist ($M^2 \leq M$).

Equivalence Relations & Partial Orders

Equivalence Relation: A relation that is *Reflexive*, *Symmetric*, and *Transitive*. It partitions set A into disjoint **Equivalence Classes** $[a] = \{x \in A \mid (a, x) \in R\}$.

Partial Order (Poset): A relation that is *Reflexive*, *Antisymmetric*, and *Transitive*.

Operations on Relations

- **Inverse Relation (R^{-1}):** $\{(b, a) \mid (a, b) \in R\}$
- **Composition ($S \circ R$):** For $R \subseteq A \times B$ and $S \subseteq B \times C$, $S \circ R = \{(a, c) \mid \exists b \in B, (a, b) \in R \wedge (b, c) \in S\}$.
- **Transitive Closure (R^*):** Smallest transitive relation containing R . Computed via Warshall's Algorithm or $R^* = \bigcup_{k=1}^n R^k$.

Recurrence Relations

An equation that recursively defines a sequence a_n in terms of previous terms.

Linear Homogeneous Recurrence with Constant Coefficients:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

Solving Technique (Characteristic Equation): Substitute $a_n = r^n$ to get $r^k - c_1 r^{k-1} - \dots - c_k = 0$.

- If roots r_1, r_2 are distinct: $a_n = \alpha_1 (r_1)^n + \alpha_2 (r_2)^n$
- If root r_1 has multiplicity 2: $a_n = (\alpha_1 + \alpha_2 n) (r_1)^n$

Exam Questions & Solutions (Relations)

PROBLEM 2.1: Equivalence Relation & Partition Proof **PROOF**

Let R be a relation on $\mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ defined by $(a, b) R (c, d) \Leftrightarrow a \cdot d = b \cdot c$. Prove that R is an equivalence relation, and describe its equivalence classes.

Proof:

1. **Reflexive:** $a \cdot b = b \cdot a$ holds for all (a, b) . Thus $(a, b) R (a, b)$.
2. **Symmetric:** Assume $(a, b) R (c, d)$. Then $ad = bc \Rightarrow cb = da \Rightarrow c \cdot b = d \cdot a$. Thus $(c, d) R (a, b)$.
3. **Transitive:** Assume $(a, b) R (c, d)$ and $(c, d) R (e, f)$.
Then $ad = bc$ and $cf = de$.
Multiply: $(ad)(cf) = (bc)(de) \Rightarrow a c d f = b c d e$.
Since $d \neq 0$ and $c \neq 0$ (if $c=0$, then $a=0$, $e=0$, holding trivially), divide by cd : $af = be$.
Thus $(a, b) R (e, f)$.

Equivalence Classes: Each class $[(a, b)]$ represents the rational number $a/b \in \mathbb{Q}$. This relation formally defines the set of Rational Numbers.

PROBLEM 2.2: Solving Non-Homogeneous Recurrence**HARD**

Solve the recurrence relation: $a_n - 5a_{n-1} + 6a_{n-2} = 4^n$ for $n \geq 2$ with initial conditions $a_0 = 1, a_1 = 4$.

Solution:**Step 1: Homogeneous Solution ($a_n^{(h)}$)**

Characteristic eq: $r^2 - 5r + 6 = 0 \Rightarrow (r - 2)(r - 3) = 0 \Rightarrow r_1 = 2, r_2 = 3$.

$$a_n^{(h)} = C_1 (2^n) + C_2 (3^n).$$

Step 2: Particular Solution ($a_n^{(p)}$)

RHS is 4^n (4 is not a root of char eq). Try $a_n^{(p)} = A \cdot 4^n$.

$$\text{Substitute into relation: } A \cdot 4^n - 5A \cdot 4^{n-1} + 6A \cdot 4^{n-2} = 4^n.$$

$$\text{Divide by } 4^{n-2}: 16A - 20A + 6A = 16 \Rightarrow 2A = 16 \Rightarrow A = 8.$$

$$\text{Thus, } a_n^{(p)} = 8 \cdot 4^n = 2 \cdot 4^{n+1}.$$

Step 3: General Solution & Initial Conditions

$$a_n = C_1 (2^n) + C_2 (3^n) + 8 \cdot 4^n.$$

$$\text{For } n = 0: C_1 + C_2 + 8 = 1 \Rightarrow C_1 + C_2 = -7.$$

$$\text{For } n = 1: 2C_1 + 3C_2 + 32 = 4 \Rightarrow 2C_1 + 3C_2 = -28.$$

$$\text{Solving yields: } C_1 = 7, C_2 = -14.$$

$$\text{Final Solution: } a_n = 7(2^n) - 14(3^n) + 8(4^n).$$

3. Functions, Types, Generating Functions & Applications

Function Classifications

A function $f: A \rightarrow B$ assigns to each element of A exactly one element of B .

- **Injective (One-to-One):** $f(a) = f(b) \Rightarrow a = b$.
- **Surjective (Onto):** $\forall y \in B, \exists x \in A$ such that $f(x) = y$ (Codomain = Range).
- **Bijjective:** Both Injective and Surjective. Invertible (f^{-1} exists).

Operations on Functions

- **Composition:** $(g \circ f)(x) = g(f(x))$ where $f: A \rightarrow B$ and $g: B \rightarrow C$.
Note: Composition is associative, but generally non-commutative ($g \circ f \neq f \circ g$).
- **Floor and Ceiling Functions:** $\lfloor x \rfloor$ (greatest integer $\leq x$), $\lceil x \rceil$ (least integer $\geq x$).

Generating Functions (GF)

A formal power series representation of a sequence $(a_0, a_1, a_2, \dots)$:

$$G(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + a_2 x^2 + \dots$$

Standard Closed Forms:

- $1 / (1 - x) = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \dots$
- $1 / (1 - ax) = \sum_{n=0}^{\infty} a^n x^n$
- $1 / (1 - x)^2 = \sum_{n=0}^{\infty} (n + 1) x^n$
- $(1 + x)^n = \sum_{k=0}^n C(n, k) x^k$ (Binomial Theorem)

Exam Questions & Solutions (Functions & GFs)

PROBLEM 3.1: Composition & Invertibility Proof PROOF

Let $f: A \rightarrow B$ and $g: B \rightarrow C$. Prove that if $g \circ f$ is injective, then f is injective.

Proof:

Suppose $g \circ f$ is injective. We need to show that $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ for any $x_1, x_2 \in A$.

Assume $f(x_1) = f(x_2)$.

Apply g to both sides: $g(f(x_1)) = g(f(x_2))$.

By definition of composition: $(g \circ f)(x_1) = (g \circ f)(x_2)$.

Since $g \circ f$ is injective, this implies $x_1 = x_2$.

Therefore, f must be injective. ■

PROBLEM 3.2: Solving Recurrence using Generating Functions HARD

Use ordinary generating functions to solve the recurrence relation: $a_n = 3a_{n-1} + 2$ for $n \geq 1$ with $a_0 = 1$.

Solution:

Step 1: Multiply by x^n and sum over $n \geq 1$:

$$\sum_{n=1}^{\infty} a_n x^n = 3 \sum_{n=1}^{\infty} a_{n-1} x^n + 2 \sum_{n=1}^{\infty} x^n$$

Step 2: Express in terms of $G(x) = \sum_{n=0}^{\infty} a_n x^n$:

LHS: $G(x) - a_0 = G(x) - 1$.

RHS: $3x G(x) + 2(x / (1 - x))$.

Step 3: Solve algebraically for $G(x)$:

$$\begin{aligned} G(x)(1 - 3x) &= 1 + (2x / (1 - x)) = (1 + x) / (1 - x) \\ G(x) &= (1 + x) / [(1 - x)(1 - 3x)] \end{aligned}$$

Step 4: Partial Fraction Decomposition:

$$(1 + x) / [(1 - x)(1 - 3x)] = A / (1 - x) + B / (1 - 3x).$$

$$1 + x = A(1 - 3x) + B(1 - x).$$

- Set $x = 1$: $2 = A(-2) \Rightarrow A = -1$.
- Set $x = 1/3$: $4/3 = B(2/3) \Rightarrow B = 2$.

Thus, $G(x) = -1 / (1 - x) + 2 / (1 - 3x) = -\sum_{n=0}^{\infty} x^n + 2 \sum_{n=0}^{\infty} 3^n x^n$.

Step 5: Extract coefficient of x^n :

$$a_n = 2 \cdot 3^n - 1.$$

4. Real-World Applications & Exam Quick Reference

Applications in Computer Science

Domain	Mathematical Concept	Application Example
Database Systems	Relational Algebra & Set Theory	SQL queries execute set operations (UNION, INTERSECT, JOIN as Cartesian product + selection).
Algorithm Analysis	Recurrence Relations	Divide-and-conquer algorithms (e.g., MergeSort $T(n) = 2T(n/2) + O(n)$ solved via Master Theorem).
Cryptography & Hashing	Functions (Injective/Hash)	Hash functions map arbitrary key spaces to array indices; collision resolution uses equivalence classes.
Combinatorics & AI	Generating Functions	Counting tree structures, integer partitions, and solving complex probabilistic pathing models.

Exam Survival Checklist

- **Matrix Powers for Relations:** Remember that $M_{R^2} = M_R \odot M_R$ (using Boolean matrix multiplication).
- **Pigeonhole Principle:** If $k + 1$ objects are placed into k boxes, at least one box contains 2+ objects.
Generalized: $\lceil N/k \rceil$ objects in at least one box.
- **Bijective Proof Strategy:** To prove f is bijection, either show direct injectivity + surjectivity OR construct an explicit inverse g and show $g \circ f = I_A$ and $f \circ g = I_B$.